

AI Research Paper Recommendation System

Software Requirements Table

The following requirements are organized for a Software Engineering / Software Requirements Specification submission.

A. Functional Requirements

ID	Requirement	Actor	Priority	Type
FR-01	User registration	Student/Researcher	High	Functional
FR-02	User login/logout	Student/Researcher	High	Functional
FR-03	Manage profile and research interests	Student/Researcher	High	Functional
FR-04	Search research papers	Student/Researcher	High	Functional
FR-05	Filter papers by author, topic, year and category	Student/Researcher	High	Functional
FR-06	View paper details	Student/Researcher	High	Functional
FR-07	Generate personalized recommendations	Student/Researcher	High	Functional
FR-08	Display recommendation explanation	Student/Researcher	Medium	Functional
FR-09	Bookmark/save papers	Student/Researcher	High	Functional
FR-10	Remove bookmarked papers	Student/Researcher	Medium	Functional
FR-11	Rate/provide feedback	Student/Researcher	Medium	Functional
FR-12	View search/recommendation history	Student/Researcher	Medium	Functional
FR-13	Manage users	Administrator	High	Functional
FR-14	Manage paper records	Administrator	High	Functional
FR-15	Validate input and display meaningful errors	All	High	Functional

B. Non-Functional Requirements

ID	Requirement	Description	Priority
NFR-01	Performance	Responsive search and recommendation behavior under expected project load	High
NFR-02	Security	Secure password hashing, authentication, and role-based authorization	High
NFR-03	Usability	Responsive, consistent, easy-to-navigate interface	High

ID	Requirement	Description	Priority
NFR-04	Reliability	Graceful handling of invalid input and common service failures	High
NFR-05	Scalability	Support growth in users, papers, and recommendation algorithms	Medium
NFR-06	Maintainability	Keep frontend, backend, recommendation logic, and data access modular	Medium
NFR-07	Privacy	Restrict profile and activity data to authorized access	High

Main data entities: User, Research Paper, Bookmark, Feedback, and History. Representative attributes include user ID, profile/interests, paper metadata, bookmark links, ratings/comments, and timestamps.